Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Connected Particles

MARKSCHEME ($g = 9.8\,\mathrm{m\,s^{-2}}$)

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1. $a = \frac{(5-3)g}{8} = 2.45$; $T = \frac{2\cdot3\cdot5\cdot g}{8} = 36.75$ **M1A1** [5]
2. $a = \frac{g}{5} = 1.96$; $T = \frac{12g}{5} = 23.52$ **M1A1** [5]
3. $a = \frac{2g}{10} = 1.96$; $T = \frac{48g}{10}$ **M1A1** [5]
4. $a = \frac{3g}{5} = 5.88$; $T = 2a = 11.76$ **M1A1** [5]
5. $a = \frac{2g}{7}$; $T = \frac{10g}{7}$ **M1A1** [5]
6. $a = \frac{g}{3}$ **M1A1** [4]
7. $a = \frac{4g}{10} = 3.92$; $v = at = 7.84$ **M1A1** [5]
8. $a = \frac{g}{5}$; $T = \frac{4g}{5}$; pulley force $2T$ (vertical strings) **M1A1** [5]
9. $a = 0$ **B1** [2]
10. Equations of motion both particles; solve for a **M1A1** [6]
11. $a = \frac{3g}{7}$; $s = \frac{1}{2}at^2$ **M1A1** [5]
12. $T = \frac{48g}{10}$ **M1A1** [4]
13. $a = \frac{3g}{5}$; $T = \frac{8g}{5}$ **M1A1** [5]
14. $a = \frac{2g}{8}$; $v^2 = 2as$ **M1A1** [5]
15. $T = \frac{3mg}{2}$ **M1A1** [4]
16–25. Form equations $T - m_2 g\sin\theta = m_2 a$ (or equiv.), $m_1 g - T = m_1 a$; solve simultaneously **M1A1** each. Standard formulae $a = \frac{(m_1-m_2)g}{m_1+m_2}$, $T = \frac{2m_1m_2g}{m_1+m_2}$ **B1**